

Printed MoS₂ memristive nanosheet networks for spiking neurons with multi-order complexity

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Artificial neurons that reproduce the rich dynamical behaviour of biological spiking are essential for neuromorphic hardware and biohybrid interfaces, yet scalable solution-processed devices with physiologically relevant spiking characteristics remain elusive. Here we demonstrate aerosol-jet-printed memristive networks of MoS₂ nanosheets that exhibit thermally activated filamentary switching and snap-back negative differential resistance, enabling volatile threshold switching in fully printed graphene/MoS₂/graphene devices on flexible substrates. In situ thermal imaging and circuit modelling reveal that current-constricted filaments formed through Joule heating govern the nonlinear switching dynamics. These printed memristors enable oscillatory and spiking neuron circuits with tunable frequencies up to 20 kHz and stable operation over more than 10⁶ cycles. Simple neuristor circuits realize first-, second- and third-order spiking complexity, including integrate-and-fire behaviour, spike latency, tonic firing, class I excitability, tonic bursting and phasic dynamics. The generated spike waveforms match physiological timescales and stimulate Purkinje neurons in mouse cerebellar slices. Our results establish printed nanosheet memristive networks as a scalable platform for bio-realistic neuromorphic hardware and flexible brain-machine interfaces.

The recent interest in data-centric workloads and biohybrid systems has highlighted the need for brain-inspired computing hardware and brain-machine interfaces^{1–5}. Limited spiking complexity in current artificial neurons necessitates spike-based neuromorphic chips to use a large number of devices for useful functionality (for example, Loihi uses 10⁶ neurons for relatively modest event-based processing)⁶. By contrast, biological systems achieve sophisticated functionality with vastly reduced numbers of neurons by exploiting adaptable, multi-order spiking complexity within each neuron⁷. For example, a

small circuit of only 30 neurons in the stomatogastric nervous system efficiently controls the complex, rhythmic motor patterns of the digestive system in crustaceans^{8,9}. Hence, mimicking the spiking complexity of biological neurons is critical to realizing the full potential of artificial neurons for bio-realistic neuromorphic computing, brain-machine interfaces and intelligent sensing^{10–12}.

Nonlinear dynamical devices have the potential to efficiently replace elaborate digital circuits as neuronal building blocks for neuromorphic computing. Nonlinear dynamical processes are represented

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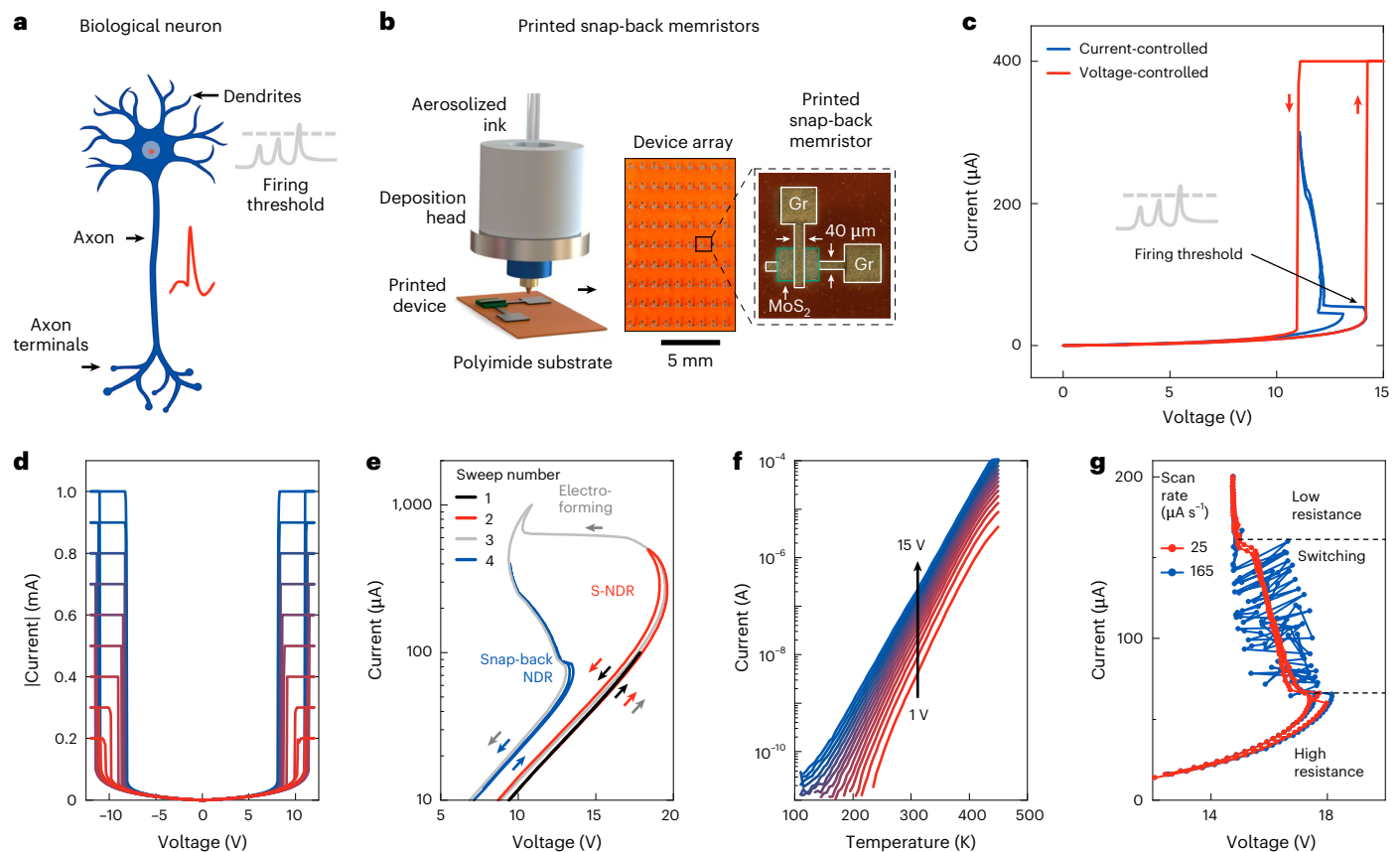


Fig. 1 | NDR in printed snap-back memristors. **a**, Schematic of a biological neuron with a firing threshold for all-or-nothing functionality. **b**, Schematic of the aerosol-jet printing process, alongside an optical image of the device array of printed snap-back memristors. **c**, Current-controlled and voltage-controlled I - V characteristics of the printed snap-back memristors, showing current-controlled snap-back NDR and voltage-controlled volatile resistive switching, mimicking the all-or-nothing functionality in biological neurons. **d**, Voltage-controlled I - V characteristics showing symmetric response for positive and negative voltages. Different plots show the I - V characteristics for different current compliance levels from 0.2 to 1.0 mA. The area of the memristive loop is a function of the

current compliance. **e**, Current-controlled characteristics before, during and after electroforming. Sweep 1 shows resistive behaviour, sweep 2 shows the onset of S-NDR, sweep 3 shows the electroforming step and sweep 4 shows the emergence of snap-back NDR. **f**, Current extracted from voltage-controlled measurements at various temperatures, showing the exponential dependence of current on temperature. **g**, Current-controlled measurements at two different scan rates, showing the presence of an unstable current window due to switching between the two resistance states. Schematic in **a** created in BioRender; Hadke, S. <https://biorender.com/n7q0h2s> (2026).

in terms of the order of complexity, which indicates the number of first-order differential equations needed to describe the evolution of distinct state variables¹³. Higher orders of complexity enable increasingly sophisticated spiking characteristics. While dynamical processes in certain transition metal oxides and organic electrochemical transistors have been used to mimic higher-order complexity spiking, they do not achieve physiologically relevant spike rates below 700 Hz and spike widths between 0.2 ms and 5 ms (Supplementary Fig. 1 and Supplementary Note 1). Moreover, for seamless integration with biological systems, it is desirable to use neuromorphic electronics on flexible substrates manufactured with solution-based printing methods, which impose additional constraints that are not met by conventional integrated circuits or memristors based on mechanically rigid, brittle materials.

Here, we use the dynamical nature of thermally activated charge transport in percolating networks of printed MoS₂-ethylcellulose (MoS₂-EC) to realize spiking neurons that closely mimic diverse biological action potentials. Specifically, thermally activated volatile threshold switching in percolating MoS₂ nanosheet networks provides current-controlled snap-back negative differential resistance (NDR), characterized by an abrupt decrease in voltage once a current threshold is reached^{14–16}. These snap-back memristors enable the realization of artificial neurons with multi-order complexity using simple circuits on

flexible substrates. The physiologically relevant spiking characteristics of our artificial neurons are validated using the resulting waveforms to stimulate the mouse cerebellar cortex. Overall, these results demonstrate the broad utility of printed snap-back memristors for bio-realistic neuromorphic computing and brain-machine interfaces.

Local activity and NDR

Biological neurons fire action potentials in response to synaptic input, often to the dendrites (Fig. 1a). While subthreshold inputs leave the neuron quiescent, suprathreshold inputs activate voltage-gated ion channels, producing an action potential that propagates down the axon. This highly nonlinear ‘all-or-nothing’ thresholding operation is a key characteristic of neuronal firing and information processing¹⁷. Emulating this thresholding functionality using conventional transistors involves elaborate circuits, such as the Izhikevich neuron, which requires 14 transistors and 2 capacitors¹⁸. Alternatively, local activity in dynamical memristive systems (defined by their ability to amplify small fluctuations in energy¹⁹) enables simpler circuits to achieve spiking with higher-order complexity. For example, a memristive neuron with neurocomputational properties similar to the Izhikevich circuit can be emulated by two memristors, two resistors and two capacitors. However, so far, these dynamic devices have relied on transition metal oxide memristors fabricated on rigid substrates, which face substantial

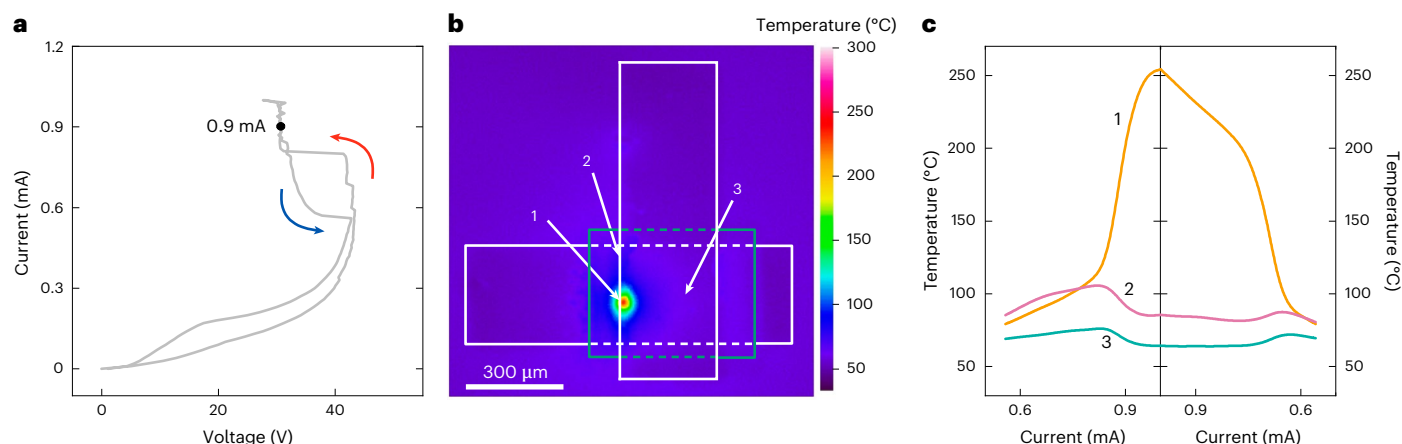


Fig. 2 | Thermal map along the snap-back NDR loop. a, Current-controlled I – V characteristics of a graphene/MoS₂/graphene snap-back memristor. **b**, Thermal map of the device at an input current of 0.9 mA showing a localized hotspot at point 1. The graphene electrodes are outlined in white, and the MoS₂ film is

outlined in green. **c**, Temperature at points 1, 2 and 3 as a function of applied current, revealing current and heat constriction from the bulk to the filament at the onset of snap-back NDR.

challenges for practical chip integration due to the specific temperature requirements of the metal–insulator transition (1,000–1,100 K for NbO₂ and 340 K for VO₂), the need for high-end cleanroom processing, and their inability to produce physiologically relevant spiking characteristics^{20–24} (Supplementary Fig. 1).

We use aerosol-jet printing of solution-exfoliated graphene and MoS₂ nanosheet inks to fabricate an array of crosspoint graphene/MoS₂/graphene devices on flexible polyimide substrates (Fig. 1b). These devices show volatile memristive behaviour in their current–voltage (I – V) characteristics. Specifically, we observe voltage-controlled (2.5 V s^{–1}) resistive threshold switching and current-controlled (25 μ A s^{–1}) snap-back NDR (Fig. 1c). The voltage-controlled mode utilizes a current compliance that not only prevents a destructive thermal runaway process but also controls the area of the memristive loop (Fig. 1d). Threshold resistive switching and snap-back NDR are the elementary physical phenomena that are used to realize the computational primitive of the neuromimetic thresholding operation^{4,25,26}.

We observe an S-type NDR (that is, a smooth transition from positive differential resistance to NDR^{14,27,28}) before electroforming and a discontinuous snap-back NDR after electroforming^{19,29,30} (Fig. 1e). Tuning the conductivity of the MoS₂-EC network by annealing at different temperatures allows control over both S-type and snap-back NDR (Extended Data Fig. 1). A strong temperature dependence of resistance is crucial for thermally activated NDR³¹. This requirement is evident from the observation that NDR is absent when using monolayer-rich MoS₂, which exhibits a weaker temperature dependence of resistance³² (Extended Data Fig. 2). Therefore, we use nanosheet networks consisting of multilayer MoS₂ flakes (Extended Data Fig. 3) to achieve optimal thermally activated NDR.

In current-controlled I – V characteristics, for a low maximum current of 100 μ A, sweep 1 shows a power law dependence ($V \propto I^{0.3}$) of voltage on the applied current without any memristive loop (Fig. 1e). Increasing the maximum current to 500 μ A reveals a smooth NDR (S-NDR) feature arising from the positive feedback between current and power dissipated by Joule heating³³. A negative temperature dependence of resistance is required for positive feedback (Fig. 1f). Successive I – V sweeps at a maximum current of 500 μ A yield I – V characteristics identical to those of sweep 2. However, increasing the maximum current to 1,000 μ A in the same device triggers an electroforming step (sweep 3), leading to snap-back NDR in subsequent I – V sweeps (sweep 4).

In accordance with other volatile memristors, the change in the I – V characteristics upon electroforming is due to the formation of a conductive filament^{14,16,33–35}. We perform thermal imaging of our printed devices (Fig. 2, Supplementary Figs. 2–4 and Supplementary Notes 2–4) to show that our results are consistent with a current constriction mechanism that was previously reported for the thermally activated snap-back NDR in TaO_x (refs. 16,36). Current constriction refers to the concentration of current density in narrow filaments due to the positive feedback between current and power dissipated by Joule heating. During the electroforming step, current constriction forms a filament along the most conductive pathway, which has a higher temperature and local conductivity than the rest of the channel, resulting in snap-back NDR. While electroforming in metal oxide memristors often leads to large device-to-device variation, the printed MoS₂ devices reported here have high yield and low variability (Supplementary Fig. 5 and Supplementary Note 5). Once the filament is formed, the device can be conceptualized as two parallel resistors, with the resistance of both being a function of the total current through the device³⁶. In this manner, two modes of current conduction are enabled: first, before current constriction, where current flows through both resistors; second, after current constriction, where the current preferentially flows through the filament due to the positive feedback between thermally activated current and Joule heating. The role of the non-filamentary region is to provide the thermal bath for current constriction to occur, and thus determines the minimum thermal energy and current threshold for switching¹⁶.

The thermal imaging results presented in Fig. 2 show the evolution of the spatially resolved temperature profile within and around the filament along the snap-back memristive loop. Figure 2a shows the current-controlled I – V characteristics, and Fig. 2b shows a thermal map of the device at an input current of 0.9 mA following the formation of a thermal filament (see Supplementary Fig. 4 for thermal maps at different input current levels). The voltage at which NDR is observed is higher in this device owing to the larger electrode size required to work within the resolution limits of the thermal imaging set-up (pixel size ≈ 5 μ m). Figure 2c shows the temperature profiles of three points (labelled 1, 2 and 3) as a function of the applied current. Point 1 corresponds to the location of the thermal filament, whereas points 2 and 3 are located away from the filament. As the current increases from 0.6 to 0.8 mA, all three points show an increase in temperature. However, as the snap-back NDR occurs and a current-constricted filament forms at a current of 0.8 mA, the temperature at point 1 abruptly

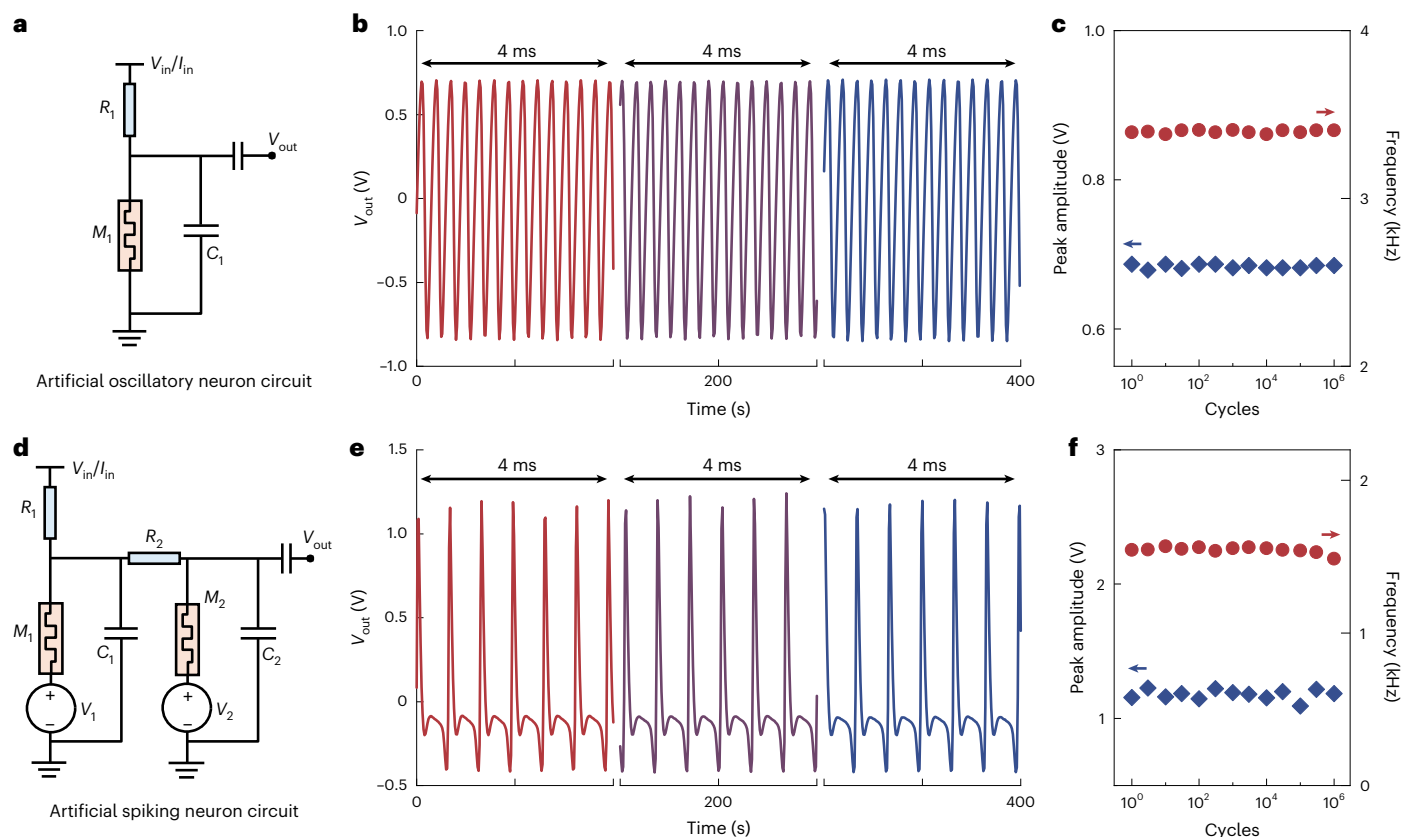


Fig. 3 | Artificial neuron circuits. **a**, Schematic of an artificial oscillatory neuron circuit. **b,c**, Endurance of artificial oscillatory neuron based on printed snap-back memristors (**b**), showing stable peak amplitude and frequency over 10^6 cycles

(c,d), Schematic of an artificial spiking neuron circuit. **e,f**, Endurance of artificial spiking neuron based on printed snap-back memristors (**e**), showing stable peak amplitude and frequency over 10^6 cycles (**f**).

increases, whereas the temperature at points 2 and 3 decreases, indicating that heat and current are constricted at the filament. The opposite trend is observed when the filament breaks in the cooling cycle at ≈ 0.7 mA, where the temperature of the filament (point 1) decreases while the temperature at points 2 and 3 increases, indicating that heat and current flow back from the filament to the bulk.

We performed time-of-flight secondary ion mass spectrometry (ToF-SIMS) to investigate localized chemical changes in the present vertical graphene/MoS₂-EC/graphene devices upon electroforming. We etched the top graphene contact with 2 keV O₂⁺ ions and then collected spatially resolved composition maps of the MoS₂-EC layer. We confirm the removal of the top graphene layer by monitoring the C⁺, C₂⁺ and C₃⁺ ions (Supplementary Fig. 6 and Supplementary Note 6). After etching the top contact, the MoS₂⁺ and MoS₂⁺ scan reveals a spatially uniform signal throughout the layer, with no evidence of a sulfur-deficient filament within the instrumental resolution of ≈ 1 μ m. However, conductive filaments can still form due to the presence of EC, as thermal annealing of MoS₂-EC has been shown to improve conductivity by facilitating electrical contact between MoS₂ nanosheets. In particular, we previously showed that the conductivity of EC-stabilized two-dimensional (2D) networks increases upon annealing, as the amorphous carbon residue from thermally decomposed EC enhances interflake charge transport³⁷. We show in Supplementary Fig. 2 that the electroforming step involves the formation of a local temperature hotspot. Hence, the electroforming step in these devices corresponds to the creation of a conductive filament with enhanced interflake transport compared with the rest of the device. As further evidence of this mechanism, we show that snap-back NDR is absent when EC is completely removed by annealing the devices at temperatures above 380 °C (Extended Data Fig. 1).

Importantly, we can reproduce the qualitative features of the I - V characteristics of our printed MoS₂ memristors using LTspice device simulations based on thermally activated filamentary switching (Supplementary Fig. 7 and Supplementary Note 7)³⁸. Specifically, we confirm that the presence of a conductive filament is sufficient to produce snap-back NDR, as well as the stochastic features in the I - V characteristics beyond the snap-back region, supporting the proposed switching mechanism. As additional evidence, we performed variable-temperature measurements and observed the expected decrease in the onset voltage of NDR with increasing temperature (Supplementary Fig. 8a and Supplementary Note 8), a characteristic of thermally activated filamentary memristors³⁴. This temperature dependence of the NDR can be qualitatively reproduced in simulations, providing additional support for our proposed model (Supplementary Fig. 8b).

As shown in Fig. 2, the current constriction in the filament sets up two conductive pathways (before and after the formation of the thermal filament). The presence of these two stable conducting pathways is supported by the observation of switching between them as a function of the applied current. In materials with both S-type NDR and snap-back NDR, current-controlled measurements show higher noise in the region between the onset of S-type NDR and the onset of snap-back NDR³⁹. We also observe this noise at a fast scan rate of 165 μ A s⁻¹ (Fig. 1g), which can be attributed to the formation and rupture of thermal filaments due to current constriction. In other words, accessing the unstable region between the high resistance state (current < 60 μ A in Fig. 1g; no current-constricted thermal filament) and low resistance state (current > 160 μ A in Fig. 1g; current-constricted thermal filament) leads to switching between these states and, hence, to noisy oscillations at fast scan rates⁴⁰ (Supplementary Fig. 7b and Supplementary Note 7).

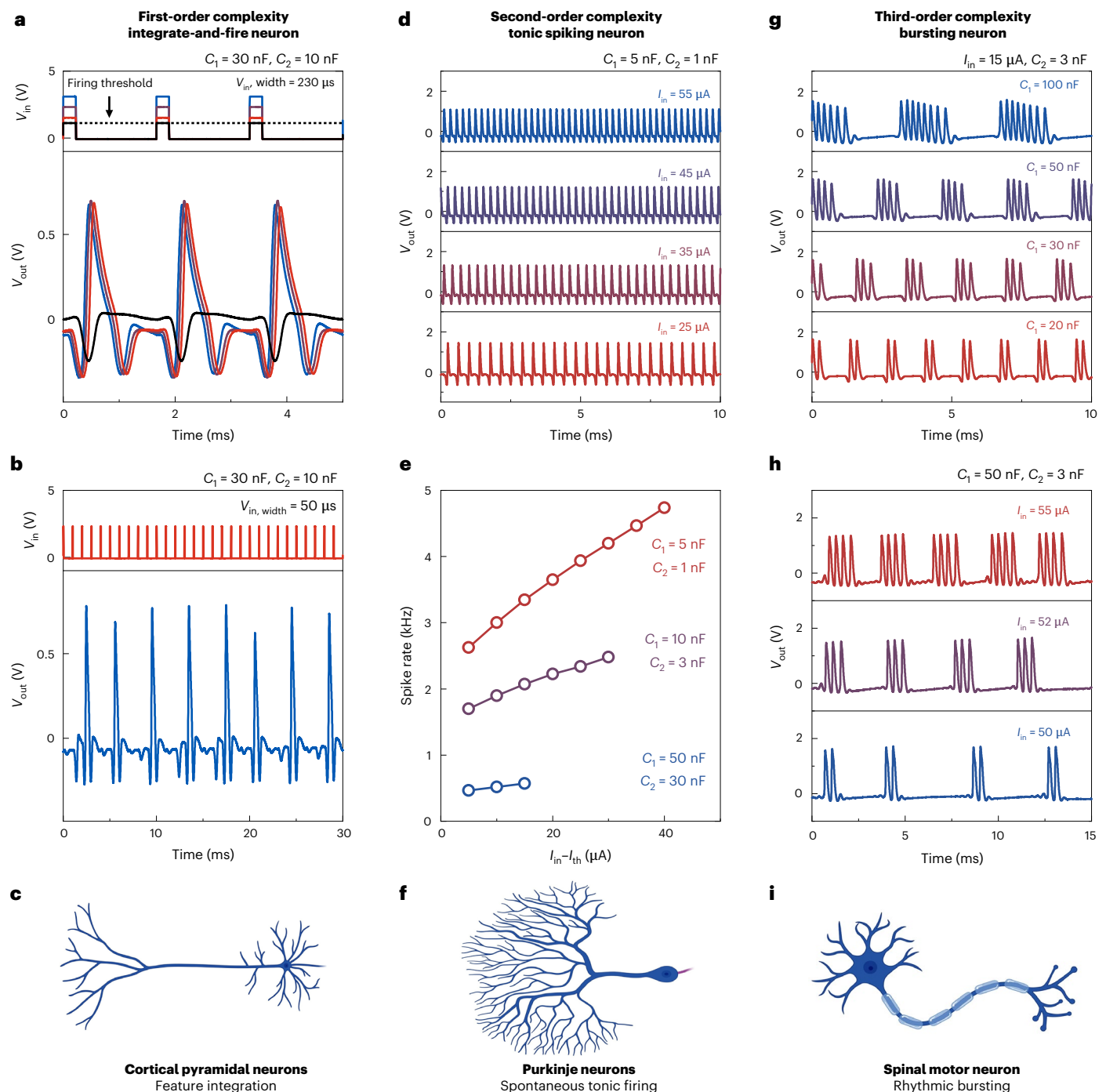


Fig. 4 | First-order, second-order and third-order complexity in spiking neuristor circuits. **a–c**, All-or-nothing (**a**) and integrate-and-fire (**b**) response in artificial spiking neurons that mimic the first-order feature integration functionality of cortical pyramidal neurons in biological systems (**c**). **d–f**, Tonic spiking (**d**) and class I excitability (**e**) in artificial spiking neurons that mimic the second-order spontaneous tonic firing functionality of Purkinje neurons in biological systems (**f**). I_{th} corresponds to the minimum current at which spiking

output waveforms are obtained. **g–i**, Bursting (**g**) and burst number adaptation (**h**) in artificial spiking neurons that mimic the third-order complexity rhythmic bursting functionality of spinal motor neurons in biological systems (**i**). The circuit shown in Fig. 3d was used to demonstrate first-order, second-order and third-order complexity spiking. Schematics in **c**, **f** and **i** created in BioRender; Hadke, S. <https://biorender.com/n7Q0h2s> (2026).

At a slow scan rate ($25 \mu\text{A s}^{-1}$), instrument averaging prevents the observation of these high-frequency oscillations, as noted previously for metal oxide memristors⁴¹.

Artificial neuron circuits

The snap-back behaviour in our devices can be used to realize oscillatory neurons (sinusoidal output waveforms with no refractory period)

based on the Pearson–Anson relaxation oscillator (Fig. 3a–c) and spiking neurons (spiking output waveforms with refractory period) based on a neuristor circuit^{26,42–44} (Fig. 3d–f). Under a constant current or voltage input, the volatile switching of the memristor charges and discharges the capacitor, leading to an oscillating output voltage (Supplementary Fig. 9 and Supplementary Note 9). The oscillations show no appreciable change in the frequency and amplitude

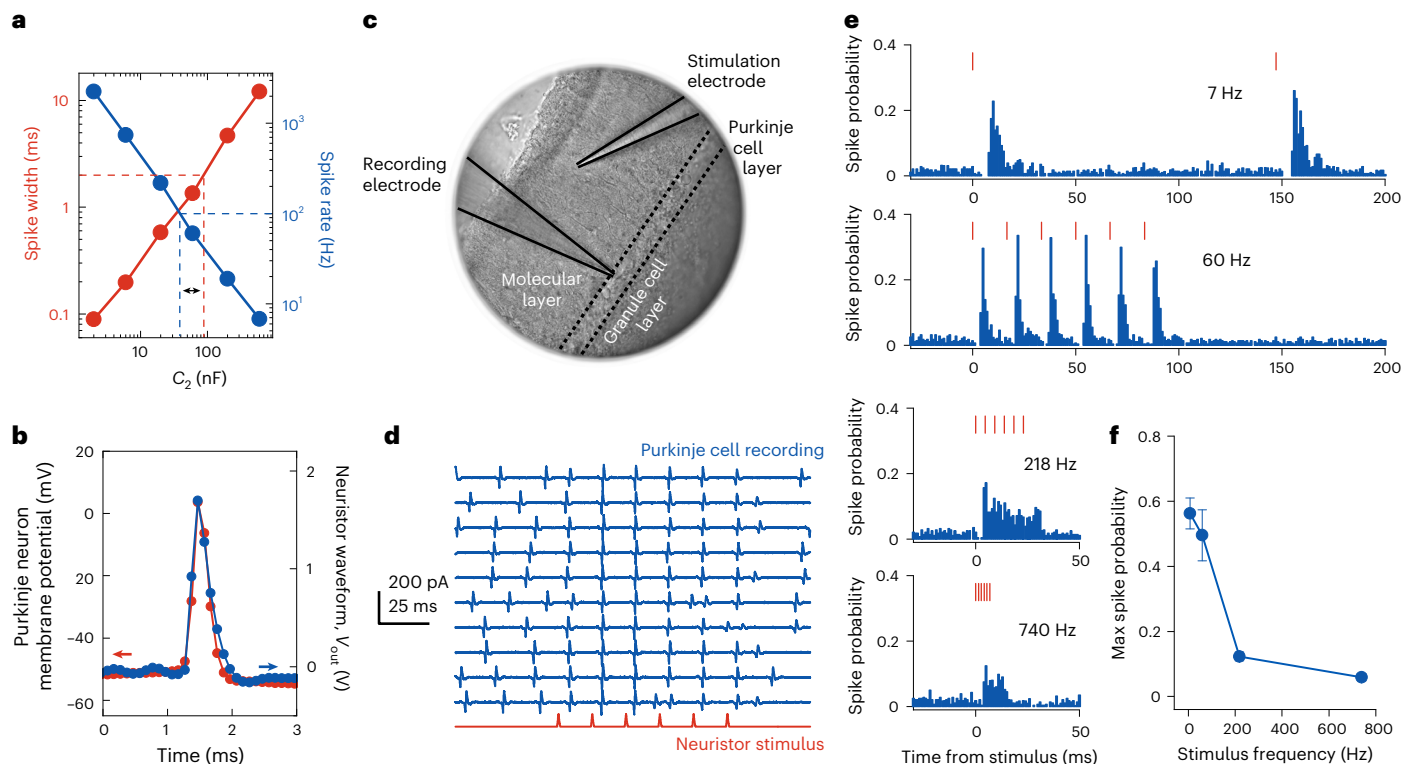


Fig. 5 | Physiologically relevant spike characteristics and in vitro cell stimulation. **a**, Dependence of spike width and spike rate on C_2 . The ratio $C_1/C_2 = 3$ was kept constant for these measurements. **b**, Comparison between neuristor circuit waveform for $C_2 = 60$ nF and action potential from a Purkinje neuron recorded in a cerebellar slice, showing the similarity in spike characteristics. **c**, Image of the experimental set-up for cell-attached recordings from Purkinje

cells combined with neuristor circuit stimulation of the parallel fibre pathway in cerebellar brain slices. **d**, Purkinje cell recording measured during parallel fibre (PF) stimulation using neuristor circuit waveforms with a spike rate of 60 Hz (ten trials shown). **e**, Mean spike probability of Purkinje cells upon neuristor circuit stimuli at all tested frequencies. **f**, Mean $\pm$ s.e.m. spike probability averaged across all stimuli and cells for each frequency ($n = 14$ cells).

over 10^6 cycles (Fig. 3b,c). The oscillator can also be triggered using light by incorporating a printed MoS₂ photoresistor as the load to realize retina-inspired sensory neurons (Extended Data Fig. 4). The frequency of these oscillations can be tuned up to 20 kHz by varying the input voltage and capacitance (Extended Data Fig. 5). Importantly, the presence of snap-back NDR is essential for high-frequency oscillations (Extended Data Fig. 6). While snap-back NDR is not required for high-frequency oscillations in oxide-based threshold-switching memristors that rely on metal–insulator phase transitions, the snap-back observed in our devices arises because the local hotspot in the filament does not form until current constriction occurs, and this constriction is responsible for the snap-back behaviour (Fig. 2 and Supplementary Fig. 2). For example, the maximum oscillation frequency is ≈ 3 Hz for S-NDR and 20 kHz for snap-back NDR. Taking advantage of this snap-back NDR, a neuristor circuit with two volatile memristors produces spiking characteristics that mimic the essential features of a biological action potential (Fig. 3e; see Supplementary Fig. 10 and Supplementary Note 10 for LTspice circuit simulations). The stability of memristive switching is demonstrated by its high endurance ($>10^6$ spikes) with no appreciable change in the frequency or peak amplitude (Fig. 3f).

Spiking neurons with multi-order complexity

We achieve first-order complexity leaky integrate-and-fire functionality using the neuristor circuit inspired by the feature integration capabilities of cortical pyramidal neurons (Fig. 4a–c). Figure 4a shows the all-or-nothing functionality of our artificial spiking neuron using the neuristor circuit in Fig. 3d, where only pulsed inputs exceeding the 1 V threshold trigger a spiking response ($R_1 = 100$ k Ω , $R_2 = 50$ k Ω , $V_1 = -11$ V and $V_2 = 13$ V). In addition, delayed spiking is also observed,

where the time delay between the input pulse and the output spike is determined by the input pulse magnitude. Figure 4a shows latencies (defined here as the time difference between the onset of the pulsed input and the peak in the spiking response) of 0.45, 0.50 and 0.55 ms, for input signals of 1.5, 2.3 and 3.1 V, respectively. The first-order complexity of integrate-and-fire is demonstrated by the integration of 2.5-V, 50- μ s pulses to trigger spiking responses (Fig. 4b; $R_1 = 100$ k Ω , $R_2 = 50$ k Ω , $V_1 = -11$ V and $V_2 = 13$ V).

Figure 4d,e shows the second-order complexity, taking inspiration from the tonic firing functionality of Purkinje neurons (Fig. 4f). Here, a constant current input leads to spontaneous tonic firing in the neuristor circuit. We show class I excitability in Fig. 4d ($R_1 = 100$ k Ω , $R_2 = 50$ k Ω , $V_1 = -13$ V and $V_2 = 14$ V), where the spike rate is modulated using the input current level. Further tunability of the spike rate is achieved by varying the capacitors in the circuit (Fig. 4e; $R_1 = 100$ k Ω , $R_2 = 50$ k Ω , $V_1 = -13$ V and $V_2 = 14$ V).

Figure 4g,h demonstrates third-order complexity with bursting behaviour that encodes the strength of the input signal, mimicking the rhythmic bursting functionality of spinal motor neurons in biological systems (Fig. 4i). In particular, the number of spikes per burst is controlled using either the capacitor C_1 (Fig. 4g; $R_1 = 100$ k Ω , $R_2 = 50$ k Ω , $V_1 = -13$ V and $V_2 = 14$ V) or the input current (Fig. 4h; $R_1 = 125$ k Ω , $R_2 = 50$ k Ω , $V_1 = -11$ V and $V_2 = 13$ V). The parallels between artificial and biological neurons are outlined here to guide the development of neuromorphic hardware for different applications. For instance, neuromorphic circuits for feature recognition may only require leaky integrate-and-fire functionality, whereas controlling the motor movements of biomimetic robots would benefit from tonic and rhythmic bursting neurons.

The neuristor circuits demonstrated above are useful not only for biomimetic computation but also for bioelectronics and

brain–machine interfaces. A key requirement for artificial neurons in brain–machine interfaces is the generation of physiologically relevant waveforms. In this context, interfacing with neuronal circuitry in the brain requires spike widths and rates within the range of neuronal signalling timescales. Here, we use the tunability in spike rate and spike width (Extended Data Fig. 7) to obtain the desired spike characteristics (Fig. 5a). Specifically, we vary the capacitances while keeping the C_1/C_2 ratio constant and measure the spike rate and spike width. With increasing capacitance, the spike width and the spike rate increase and decrease, respectively. For C_2 in the range of 60–100 nF, we obtain physiologically relevant spiking characteristics. In Fig. 5b, we show the similarity between the neuristor circuit output waveform and the action potential of a cerebellar Purkinje neuron from a mouse brain. In addition to tonic spiking, we also show phasic spiking with bio-realistic spike characteristics (Extended Data Fig. 8).

In vitro validation with biological neurons

To validate the physiological relevance of our spike waveforms, we explored their ability to activate neuronal circuits in vitro in brain slices from the mouse cerebellum⁴⁵ (Fig. 5c). We performed extracellular recordings from the principal neurons of the cerebellar cortex (that is, Purkinje cells) while stimulating their primary source of excitatory input (that is, parallel fibres) with our neuristor circuit waveforms (Fig. 5d). A series of four waveforms were tested that varied concurrently in spike width (5.9, 0.7, 0.3 and 0.1 ms) and frequency (7, 60, 218 and 740 Hz). To quantify Purkinje cell responses, the time of action potentials relative to stimulation onset were collected in 1-ms binned histograms, which were converted into spike probability by dividing by the trial count (mean $\pm$ s.d.: 20 ± 2 trials per condition per cell) and then averaged across cells ($N = 14$ cells tested with all conditions; Fig. 5e). All waveforms reliably elicited Purkinje cell spiking above their basal firing probability, consistent with parallel fibre excitation (all $P < 0.01$; Fig. 5e). The largest increase in spike probability occurred for waveforms with spike rates below 200 Hz (compared with 218 Hz and 740 Hz; all $P < 0.001$; Fig. 5f), indicating the limits of the physiologically relevant parameter space. Between the 7-Hz and 60-Hz waveforms, with pulse widths of 5.9 ms and 0.7 ms, respectively, the latter is preferred, as shorter pulses reduce total charge delivery and minimize the risk of thermal damage to the cells. Although stronger responses to lower frequencies could result from multiple factors, the primary reason is that higher frequencies (218 Hz and 740 Hz) exceed the maximum firing rate of Purkinje cells (≈ 200 Hz), making them less effective. Taken together, we demonstrate that our neuristor circuit can produce waveforms that operate within the activity regime of biological circuits, inducing synaptic release and subsequent action potential firing in postsynaptic neurons (Fig. 5e,f). The results pave the way for implementing printed neuristor circuits in bio-realistic brain–computer interfaces.

Conclusion

We have demonstrated that thermally activated filamentary switching-induced snap-back NDR in fully printed graphene/MoS₂/graphene volatile memristors enables the realization of multi-order complexity spiking artificial neurons. This observation of purely thermal snap-back NDR expands the range of materials available for such functionalities, moving beyond the metal oxide counterparts that require metal-insulator transitions in Mott insulators to exhibit snap-back behaviour. The ability to print solution-processed 2D nanosheets on arbitrary substrates also opens possibilities to fabricate bio-realistic hardware for neuromorphic computing, brain–machine interfaces and related bioelectronic applications. We have further demonstrated a retina-inspired sensory neuron that encodes the strength of the light stimulus in the action potential spiking rate. The ability to tune our neuristor circuit action potentials with external circuit elements implies that this base design can be generalized to build sensory neurons for additional types of stimuli by replacing the photoresistor with other

resistive sensors to realize multimodal sensing artificial neurons for intelligent sensing systems^{40,46}. The operating voltage of the devices can be reduced by using materials and percolating networks with a stronger dependence of resistance on temperature. While the memristive behaviour reported here relies on the interflake transport between MoS₂ nanosheets, this phenomenon can probably be realized in percolating networks of other 2D materials, such as TaS₂, which also show thermally activated phase transitions, which can potentially be used to obtain even higher complexity thresholding behaviour with multiple NDR regions^{25,47,48}. Finally, the physiologically relevant spiking characteristics of our artificial neurons make them suitable for applications in neuroprosthetics and bionic devices. Because neurons are one of the basic building blocks of brain-inspired biomimetic computation, the solution-processed and printable nature of our devices has the potential to have a similar impact in the field of neuromorphic electronics that printed transistors have enabled in flexible and wearable electronics⁴⁹.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41565-026-02149-6>.

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Methods

Ink preparation and device fabrication

MoS₂ was exfoliated using shear mixing. A dispersion of 120 g MoS₂ powder, 8 g EC and 800 g ethanol was shear mixed (Silverson L5M-A) at 10,000 rpm for 2 h. The dispersion was then centrifuged at a g-force of 2,000g for 30 min to remove poorly exfoliated MoS₂. The supernatant (500 g) was flocculated using 4% NaCl (300 g) at 7,500 rpm for 6 min. The dispersion was then filtered, washed with deionized water and dried to obtain MoS₂-EC powder. Printable MoS₂-EC inks were then prepared using this powder by bath sonicating a 10 mg ml⁻¹ dispersion in ethanol for 8 h and filtering through a 0.7-μm syringe filter. Finally, 10% (by volume) of terpineol was added to the ink before printing. Monolayer-rich MoS₂ ink was prepared for photoresistors using the previously reported megasonic exfoliation of electrochemically intercalated MoS₂ (ref. 32).

For the printed graphene electrodes, graphite powder (≈40 mg ml⁻¹) was dispersed into an ethanol solution containing (≈4 mg ml⁻¹) EC. The solution was shear mixed using a continuous-flow inline shear mixer (Silverson Machines 200L). The resulting dispersion was then centrifuged with a g-force of ≈8,100g for 30 min, followed by a flocculation step in which the supernatant was combined at 1.74 w/w with a ≈40 mg ml⁻¹ NaCl solution and centrifuged once more at ≈9,380g for 7 min. The sediment was collected and rinsed with deionized water under vacuum filtration before being dried with an infrared lamp. Printable graphene-EC inks were then prepared using the same solvent formulation as the printable MoS₂-EC inks.

Crosspoint graphene/MoS₂/graphene devices were printed using aerosol-jet printing (Optomec AJ-200) by sequentially printing the bottom graphene contact, followed by the MoS₂ channel and finally the top graphene contact. Both the graphene-EC and MoS₂-EC inks were printed using a 150-μm nozzle, a sheath flow rate of 50 sccm, a carrier gas flow rate of 12 sccm and a printing speed of 1 mm s⁻¹. The ink and stage temperatures were 30 °C and 60 °C, respectively. The typical thicknesses of the graphene and MoS₂ films were 300 nm and 2.5 μm, respectively, as determined using a confocal microscope. The thickness of the films was controlled through the number of printing passes. Photoresistors were printed using the inks prepared from electrochemically exfoliated MoS₂ with printed graphene electrodes. The photoresistors and the memristors were annealed in Ar atmosphere at 350 °C for 2 h to remove the EC.

Materials characterization

Raman spectra were acquired using a Horiba Xplora Raman/PL system with a 532-nm laser with a spot size of ≈1 μm². Optical absorption spectra were acquired using a Cary-5000 (Agilent) ultraviolet-visible spectrophotometer. A plastic cuvette with a path length of 1 cm was used for these measurements. Atomic force microscopy (Asylum Cypher) was performed in tapping mode with NanoWorld NCHR-W Si cantilevers to obtain flake size and thickness distributions. The thickness of the printed films was measured using an Olympus (LEXT OLS5000) three-dimensional laser scanning confocal microscope. In situ thermal imaging of vertical graphene/MoS₂/graphene memristors was performed in ambient using an Optotherm Micro Thermal Imaging System. Thermal maps were captured as video files, and frames corresponding to *I*-*V* characteristics were extracted using FFmpeg (<https://ffmpeg.org/>). ToF-SIMS was performed using an ION TOF M6 system. Before the measurement, the sample was coated with a 10-nm osmium film from sputtering to reduce charging issues from the insulating polyimide substrate. Ion beam etching to remove the precoated osmium layer and the top graphite contact from the area of interest was performed using O₂⁺ sputtering (2 keV, 600 nA). Measurements were performed using a 15 keV Bi₁⁺ primary ion at a maximum pressure of 2 × 10⁻⁸ mbar. Charge compensation was used to further reduce charging effects during the measurement. The imaging was performed by sawtooth raster scan over an area

of 500 × 500 μm² at 2,048 × 2,048 pixels, with the detector set to monitor negative ions. The acquired mass spectra were analysed with SurfaceLab 7 software.

Device and circuit measurements

Electrical measurements were performed in a probe station (LakeShore CRX 4K) equipped with a diode laser (LP520MF100, Thor Labs). Laser calibration was performed using a Si photodiode (S120C, Thor Labs) and energy meter (PM100D, Thor Labs). Variable temperature measurements were performed in the same set-up using a closed-cycle He cryostat. The measurements were performed in a vacuum at a pressure less than 10⁻⁴ torr. Current-voltage characteristics were measured using a Keithley SCS-4200 parameter analyser. Source-meter units (Keithley 2400) were used as constant current and voltage sources. The oscillator and neuristor circuits were assembled on a breadboard to connect the printed devices to the measurement apparatus using external resistors and capacitors. The output waveforms were measured using a Tektronix MDO4104C oscilloscope, while the pulsed voltage inputs were generated using an Agilent 33500B waveform generator. Circuit simulations were performed using LTspice, following a previously reported model³⁸. The memristive element was modelled as a voltage-dependent current source, with a resistance equal to the state-dependent resistance equation. Then, the model parameters for the memristor were varied to replicate the current-voltage curve measured experimentally.

In vitro experiments

Experimental model and study participant details. All experiments on mice were conducted in accordance with institutional guidelines using methods approved and overseen by the Northwestern University IACUC. All mice were housed with standard care in an accredited veterinary facility, temperature range 21.1–23.3 °C and 30–70% humidity, on a standard 12-h–12-h day–light cycle, with free access to food and water. Experiments were performed on brain slices from wild-type C57BL/6J mice (*N* = 2; P21–32, Jackson Labs). Both sexes were used.

Electrophysiology and neuristor circuit stimulation. Mice were deeply anaesthetized with isoflurane and decapitated. All salts were from Sigma-Aldrich. Cerebella were sliced parasagittally in warm (35 °C) artificial cerebrospinal fluid containing (in mM) 123 NaCl, 3.5 KCl, 1.25 NaH₂PO₄, 26 NaHCO₃, 1 MgCl₂, 1.5 CaCl₂ and 10 D-glucose and bubbled with 95% O₂/5% CO₂ (carbogen). Slices recovered for > 1 h in carbogenated artificial cerebrospinal fluid (22–25 °C). Extracellular loose cell-attached recordings of Purkinje cell action potentials were made in cerebellar brain slices with borosilicate pipettes (3–6 MΩ) filled with Tyrode's solution (in mM: 150 NaCl, 4 KCl, 2 CaCl₂, 2 MgCl₂, 10 HEPES and 10 glucose, pH 7.35 with NaOH) at room temperature (21 °C). Recordings were made with a MultiClamp 700B amplifier, digitized and acquired with pClamp software (Molecular Devices). The peak voltage of all neuristor circuit signals was set to 2.4 V by outputting the signal through a digital-to-analogue converter and subsequently passing it through a constant current source, which ensured that the stimulated Purkinje cells received the same peak amplitude of current per neuristor circuit action potential, independent of the spike waveform and frequency. To stimulate parallel fibre inputs of recorded Purkinje cells, positively thresholded neuristor circuit waveforms were injected into the molecular layer. Peak injected currents were adjusted to be within the range of 10–50 μA. The response of each cell was tested over 20 trials for each frequency (740, 218, 60 and 7 Hz). Histograms of spike times for each cell were generated with 1-ms bins and were converted to spike probability by dividing by the number of trials. Spike probability histograms from individual cells were averaged together to generate mean spike probability histograms. Maximum spike probabilities were measured within each interstimulus interval. All data were analysed in MATLAB.

Statistics and reproducibility

The full set of neuristor stimulation experiments was repeated in all recorded Purkinje cells ($n = 14$). Purkinje cells were included in analyses if they had at least ten trials per condition (mean $\pm$ s.d.: 20 ± 2 trials per condition per cell). Data collection and analysis were not performed blind to the conditions of the experiments, and the experiments were not randomized. Data are reported as mean $\pm$ standard error of the mean (s.e.m.) unless noted as mean $\pm$ s.d. Statistical significance was assessed with paired Student's t -tests. Normality was assessed and confirmed with an Anderson–Darling test. No statistical methods were used to predetermine sample sizes.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All the data that support the findings of this study are available within the Article and its Supplementary Information. Source data are provided with this paper.

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Author contributions

S.S.H., V.K.S. and M.C.H. conceived the idea and designed the experiments. S.S.H., S.D.-A., S.L. and J.H.L. produced the printable inks. S.S.H., C.N.K., L.L. and A.C. printed the devices. S.S.H. and X.Z. performed the circuit simulations. S.S.H. performed the electrical measurements. S.S.H. and S.D.-A. measured the absorption and Raman spectra. S.D.-A. measured the flake characteristics using AFM. M.I.B.U. performed ToF-SIMS measurements. S.T.B. and M.H. designed and performed biological experiments under the supervision of I.M.R. All authors wrote the manuscript and discussed the results.

Competing interests

The authors declare no competing interests.

Additional information

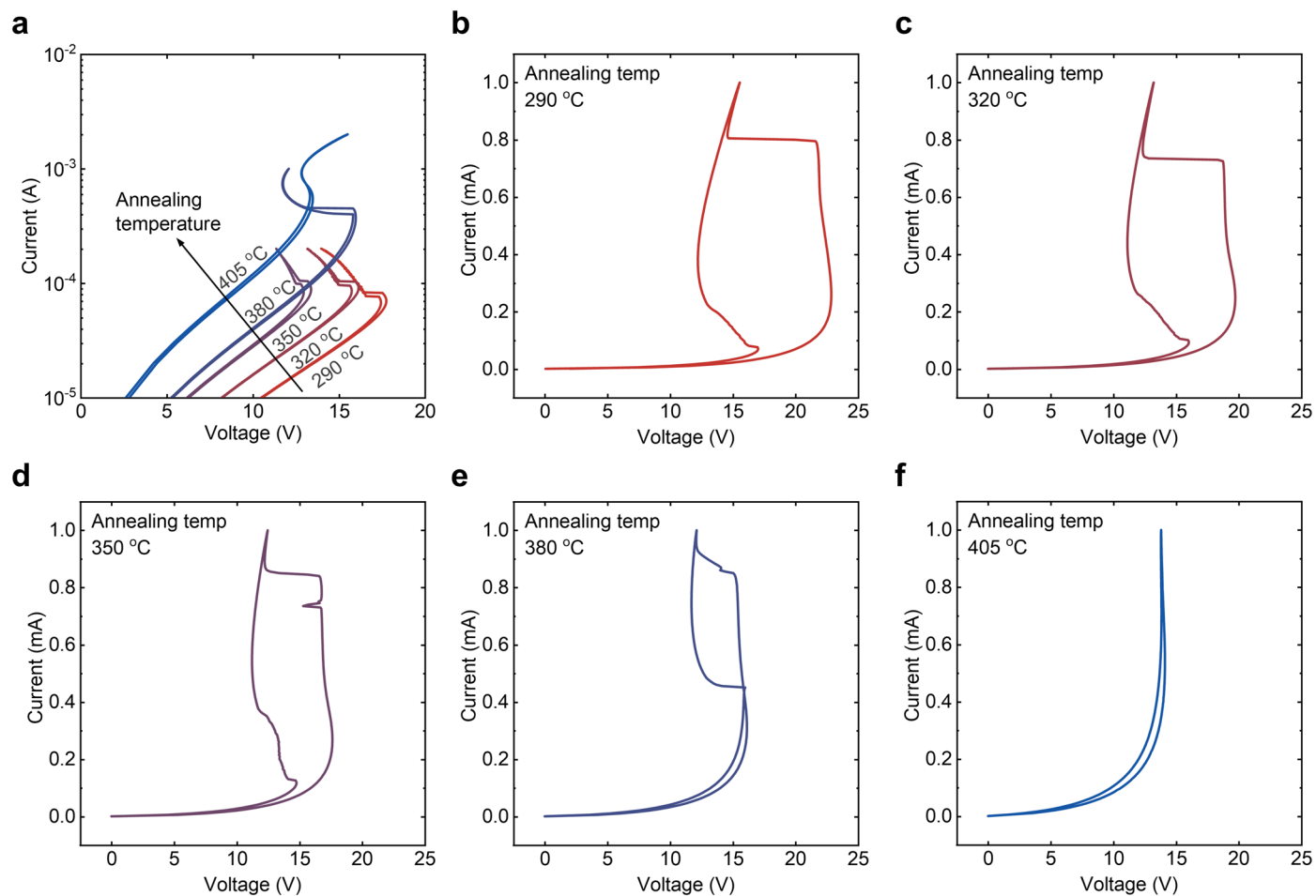
Extended data is available for this paper at <https://doi.org/10.1038/s41565-026-02149-6>.

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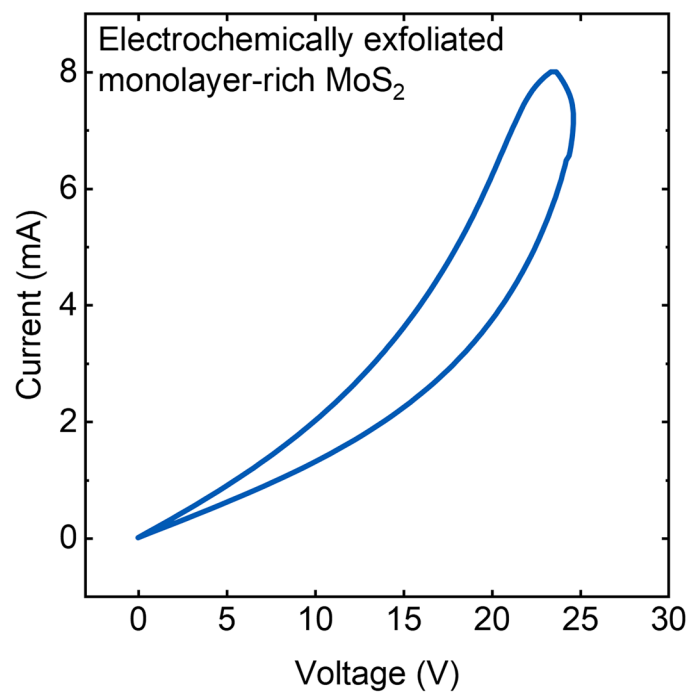
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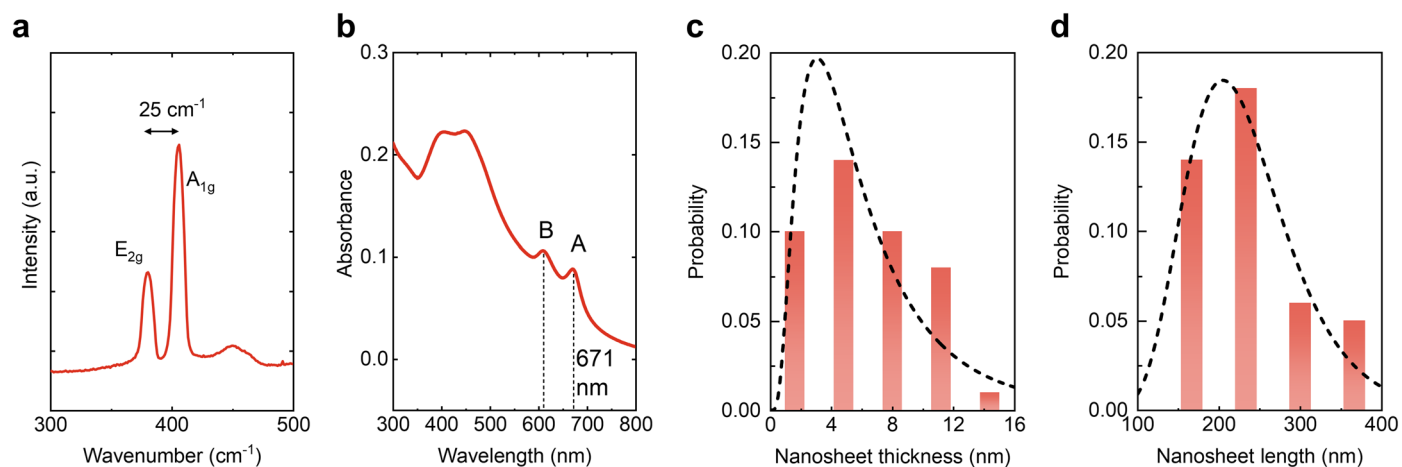


Extended Data Fig. 1 | Effect of annealing temperature on S-NDR and snap-back NDR. (a) Current-controlled I - V characteristics showing the effect of annealing temperature on the snap-back behavior. A transition from snap-back NDR to S-NDR is observed at higher annealing temperatures. (b-f) Current-controlled I - V characteristics during the electroforming step,

showing that the post-electroforming characteristics depend on filament formation during electroforming. For the sample annealed at 405 °C, electroforming does not lead to snap-back NDR, highlighting that a higher limit of conductance exists above which snap-back NDR is not observed in these graphene/MoS₂-EC/graphene devices.

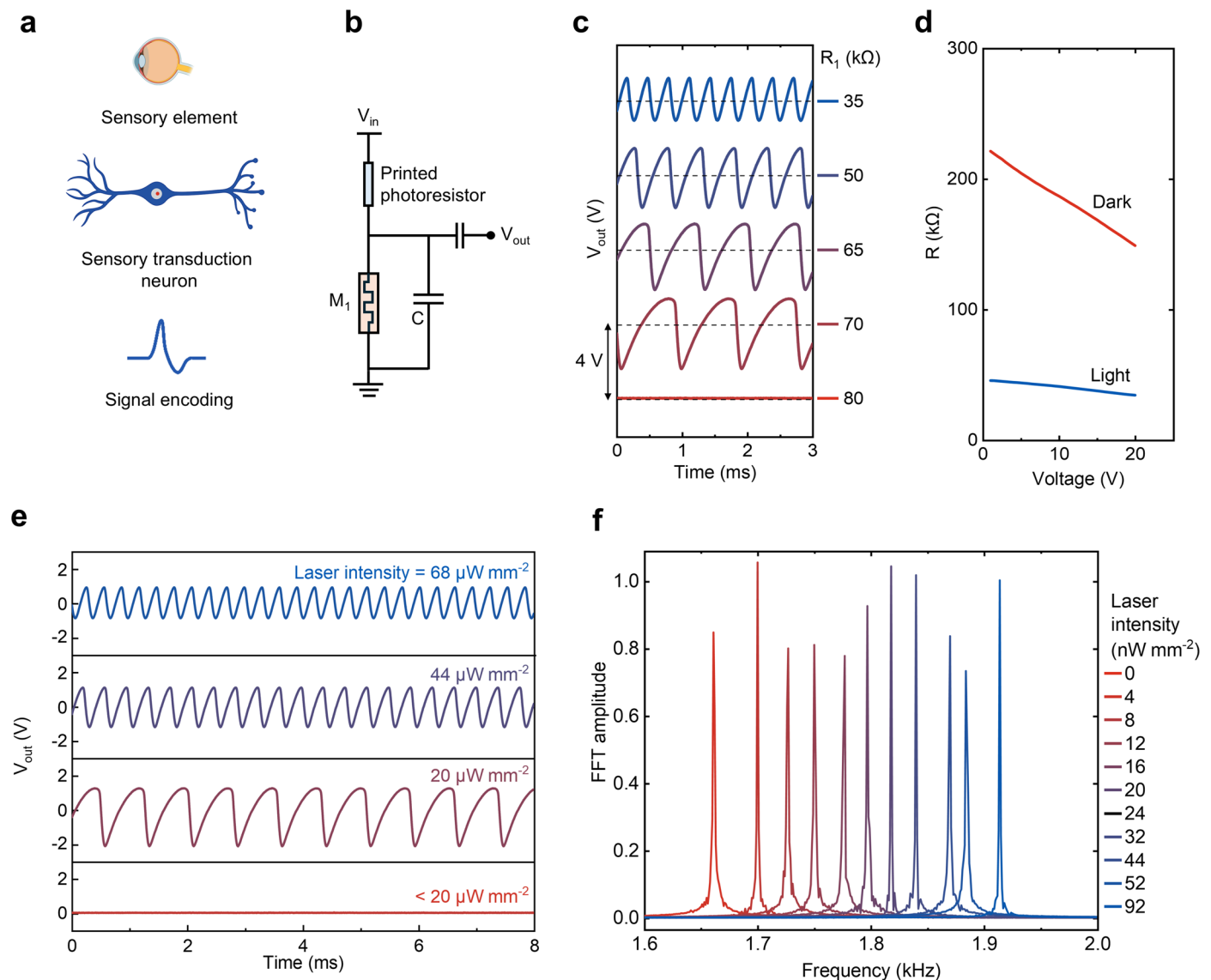


Extended Data Fig. 2 | Monolayer-rich MoS₂. Voltage-controlled *I*-*V* characteristics of electrochemically exfoliated monolayer-rich MoS₂ showing the absence of NDR features.



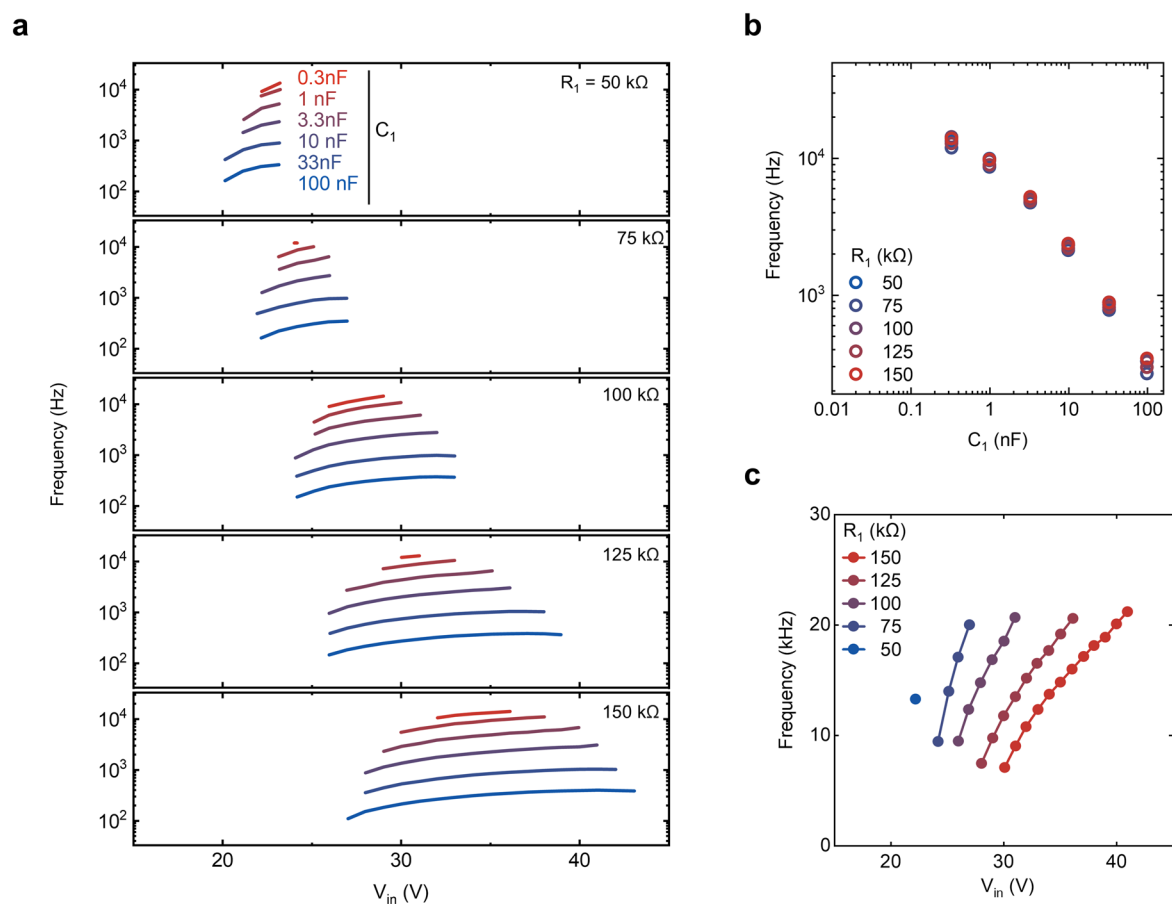
Extended Data Fig. 3 | Characteristics of shear exfoliated multilayer MoS₂ flakes. (a) Raman spectrum of shear exfoliated MoS₂. (b) Absorption spectrum of shear exfoliated MoS₂. The A-exciton peak position indicates the presence

of multi-layer MoS₂ flakes. (c) Nanosheet thickness and (d) length determined from AFM measurements of shear exfoliated MoS₂ flakes (a total of 43 flakes were measured). The dashed lines are log-normal fits.

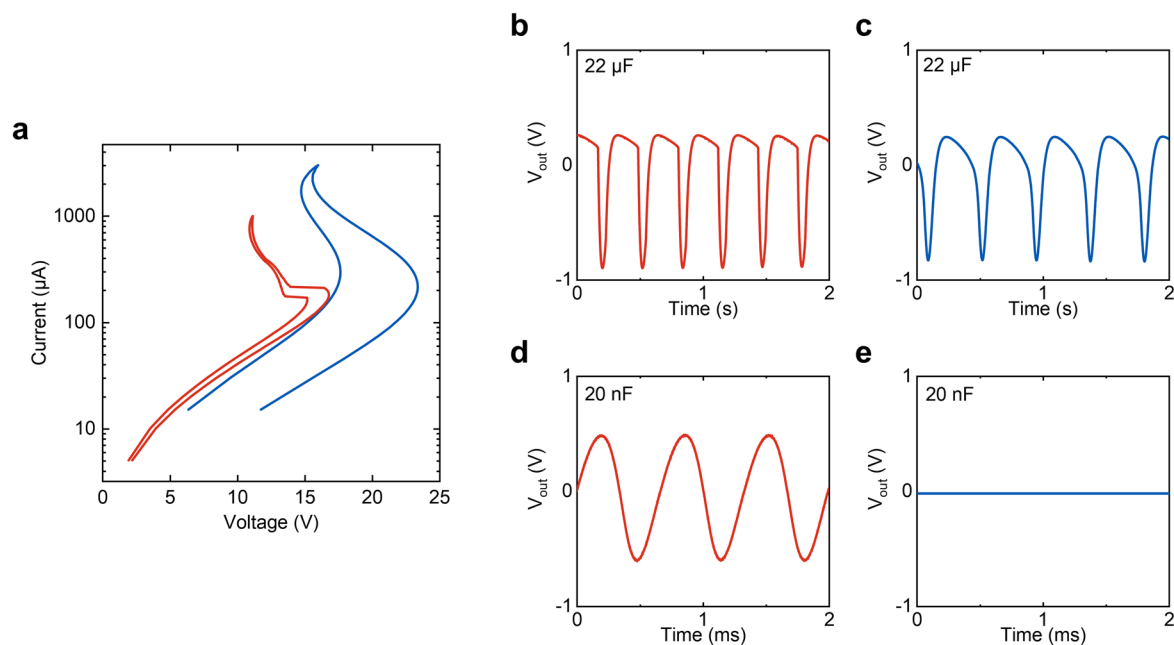


Extended Data Fig. 4 | Retina inspired oscillatory neuron. (a) Schematic showing a sensory transduction neuron that encodes the strength of input signals into the spike rate (frequency of action potentials) generated in the axon. (b) Schematic of a retina-inspired light-sensitive oscillatory neuron using a printed MoS₂ photoresistor as a sensory element together with a printed MoS₂ volatile memristor and a capacitor to generate tunable oscillatory responses. The load line resistance is determined by the photoresistor, which triggers and modulates the response of the artificial neuron. The voltage drop across the photoresistor acts as the input to the capacitor and the memristor, and is used to modulate the oscillatory dynamics of the circuit. (c) Effect of load resistor on the oscillator circuit to determine the change in resistance required to enable retina-inspired operation ($V_{in} = 22$ V and $C_1 = 10$ nF). A resistance < 80 k Ω is needed to achieve self-sustained oscillations. (d) Resistance under dark and illumination

of a printed MoS₂ photoresistor that achieves a resistance > 80 k Ω in the dark and < 80 k Ω under 516 nm illumination. (e) Integration of the photoresistor with an oscillator circuit to enable retina-inspired signal transduction. For light intensity $< 20 \mu\text{W mm}^{-2}$, the oscillatory neuron is in its resting state (the value of this threshold can be tuned using V_{in}). When the light intensity crosses the predetermined threshold, the artificial oscillatory neuron fires, following the all-or-nothing characteristics of biological neurons. The intensity of light is captured in the frequency of the oscillations. (f) Fast Fourier Transforms (FFT) of oscillations under different illumination intensities, showing that the oscillatory neuron is sensitive enough to capture small changes (4 nW mm^{-2}) in light intensity. Schematic in a created in BioRender; Hadke, S. <https://biorender.com/n7q0h2s> (2026).



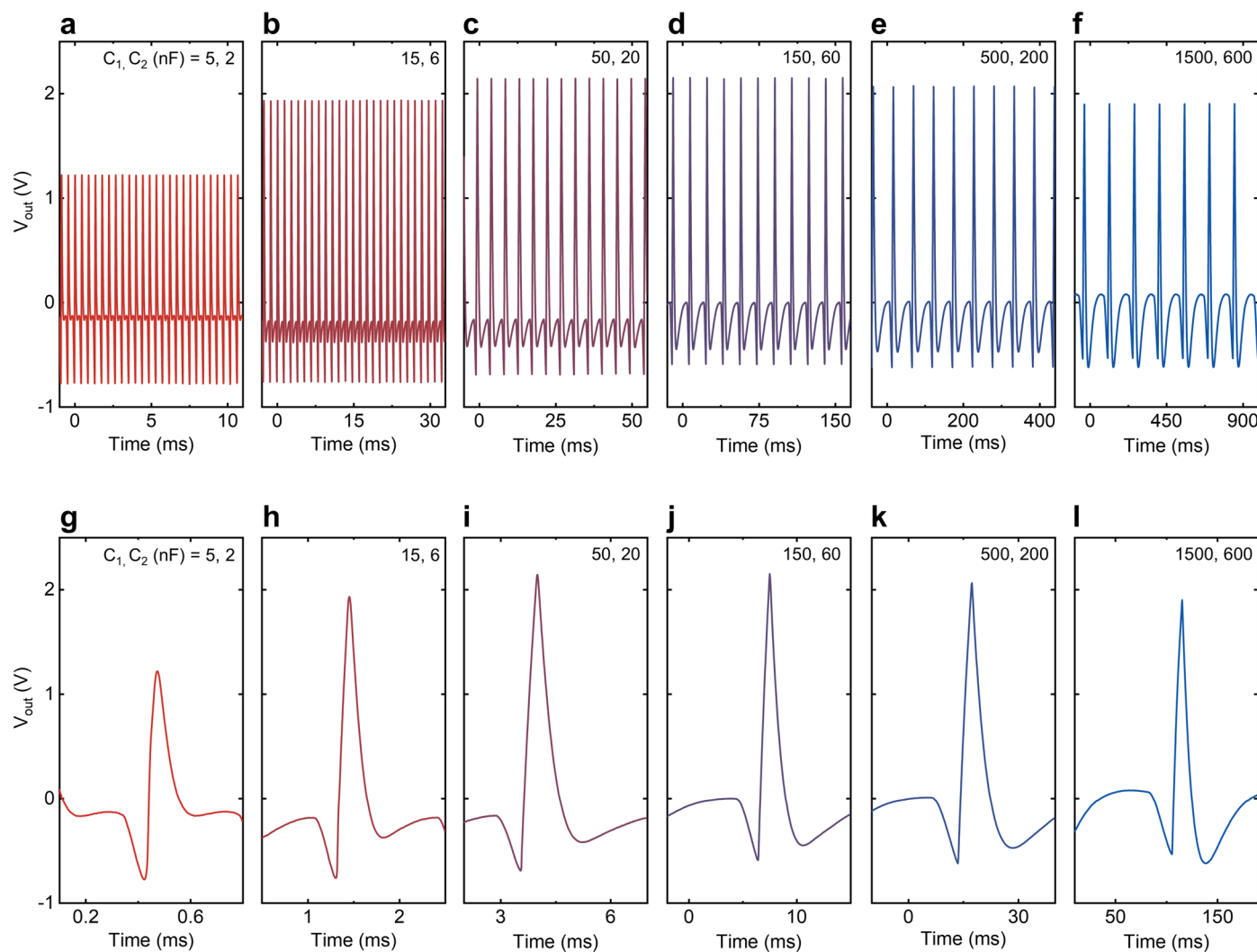
Extended Data Fig. 5 | Oscillatory neuron tunability. (a) Oscillation frequency of the circuit in Fig. 3a as a function of V_{in} for different values of R_1 and C_1 . (b) Oscillation frequency as a function of C_1 for different values of R_1 . (c) Oscillation frequency as a function of V_{in} for different values of R_1 in the absence of C_1 . In this case, the parasitic capacitance drives the oscillations.



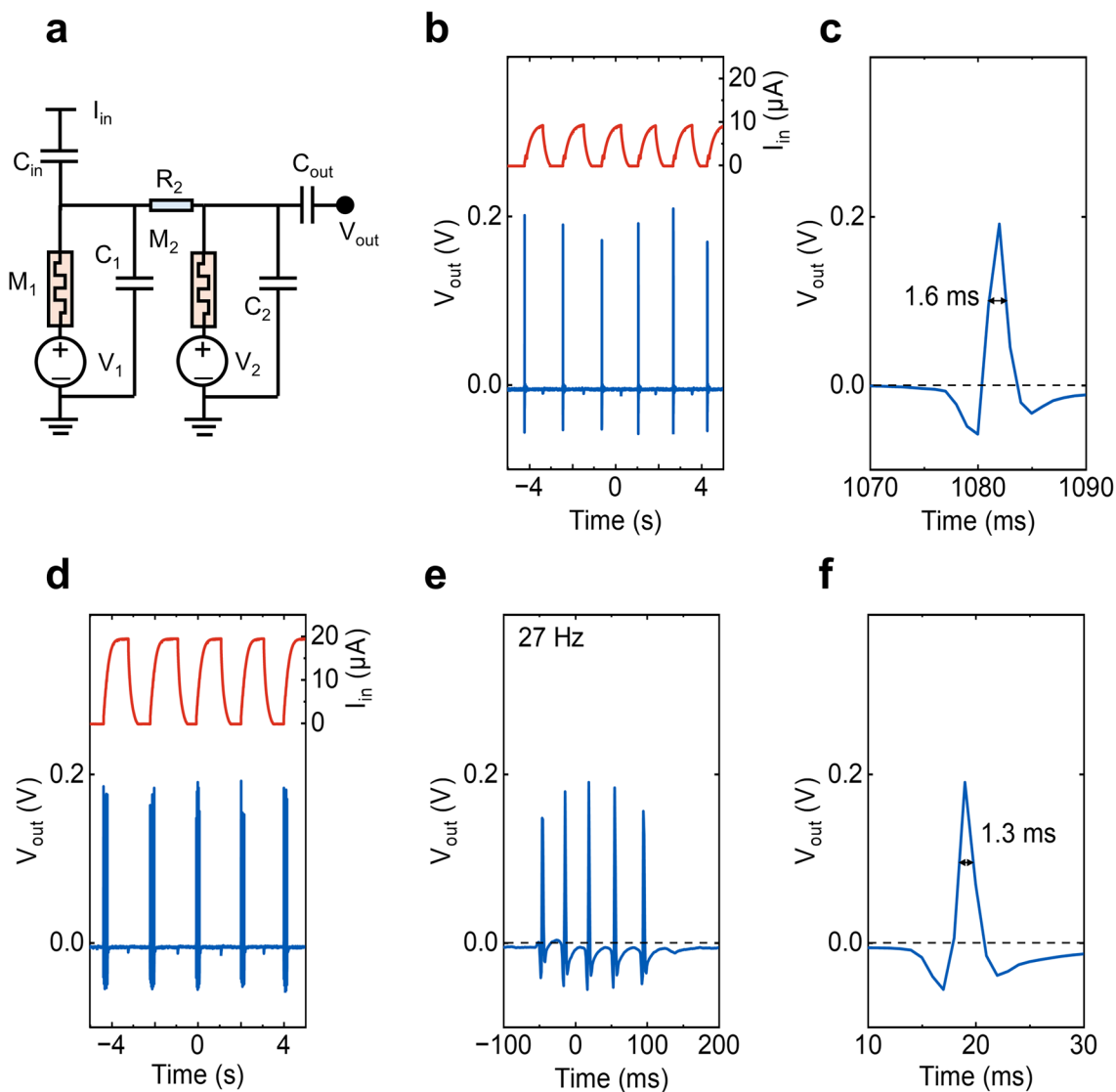
Extended Data Fig. 6 | Snap-back NDR and high frequency oscillations.

(a) Current-controlled I - V characteristics of two memristors showing snap-back NDR and S-NDR. Response of the oscillatory neuron circuit for $C_1 = 22 \mu\text{F}$, $R_1 = 40 \text{ k}\Omega$, using (b) snap-back NDR ($V_{\text{in}} = 26 \text{ V}$) and (c) S-NDR ($V_{\text{in}} = 37 \text{ V}$) devices, showing oscillatory behavior in both cases. Response of the oscillatory neuron

circuit for $C_1 = 20 \text{ nF}$ using (d) snap-back NDR ($V_{\text{in}} = 26 \text{ V}$) and (e) S-NDR ($V_{\text{in}} = 30 \text{ to } 50 \text{ V}$) devices, showing oscillatory behavior only for the snap-back NDR case. These results indicate that the presence of snap-back NDR is essential to achieve high oscillation frequencies.



Extended Data Fig. 7 | Spiking neuron tunability. (a–f) Response of the artificial spiking neuron circuit for different values of C_1 and C_2 at $I_{in} = 15 \mu A$. (g–l) Magnified views of spike waveforms in a–f, showing individual spikes. Here, the circuit shown in Fig. 3d was used with the parameters $V_{in} = 2$ V, $R_1 = 100$ k Ω , $R_2 = 50$ k Ω , $V_1 = -12$ V, and $V_2 = 14$ V.



Extended Data Fig. 8 | Phasic spiking and bursting. (a) Schematic of the circuit diagram for the phasic spiking and bursting neuristor circuits. Here, $C_{in} = 200$ nF, $C_1 = 1.5$ μ F, $C_2 = 200$ nF, $R_2 = 50$ k Ω , $V_1 = -12$ V, and $V_2 = 15$ V. (b) Response of the circuit for pulsed current inputs, showing phasic spiking behavior.

(c) Magnified view of the waveform in (b), showing physiologically relevant spike characteristics. (d) Response of the circuit for pulsed current inputs, showing phasic bursting behavior. (e-f) Magnified view of the waveform in (d), showing physiologically relevant spike characteristics.

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Software and code

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Data collection Keithley Test Environment Interactive (KTEI) V9.0, Cary WinUV, LabSpec 6, and Thermalzy were used for data collection.

Data analysis MATLAB R2023a, OriginPro 2023, Gwyddion 2.62, SurfaceLab 7, and FFmpeg were used for data analysis.

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☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

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☒	☐ ChIP-seq
☒	☐ Flow cytometry
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Laboratory animals	Experiments were performed on brain slices from wild-type C57BL6/J mice (P21-32, Jackson Labs).
Wild animals	Study did not involve wild animals.
Reporting on sex	Both sexes were used and all data was pooled.
Field-collected samples	Study did not involve samples collected from the field.
Ethics oversight	All experiments were conducted in accordance with institutional guidelines with methods approved and overseen by the Northwestern University IACUC.

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Plants

Seed stocks	Study did not involve seed stocks.
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Authentication	Study did not involve plants.